As a matter of fact, this *is* what most physicists have done since the early decades of the twentieth century, and there's no denying that it's been a distinctly fruitful approach. The progress science has made by working in this disjointed framework is impressive. All the same, there are a number of reasons why the antagonism between general relativity and quantum mechanics must be reconciled. Here are two.

First, at a gut level, it is hard to believe that the deepest understanding of the universe consists of an uneasy union between two powerful theoretical frameworks that are mutually incompatible. It's not as though the universe comes equipped with a line in the sand separating things that are properly described by quantum mechanics from things properly described by general relativity. Dividing the universe into two separate realms seems both artificial and clumsy. To many, this is evidence that there must be a deeper, unified truth that overcomes the rift between general relativity and quantum mechanics and that can be applied to *everything.*We have one universe and therefore, many strongly believe, we should have one theory.

Second, although most things are either big and heavy or small and light, and therefore, as a practical matter, can be described using general relativity *or* quantum mechanics, this is not true of all things. Black holes provide a good example. According to general relativity, all the matter that makes up a black hole is crushed together at a single minuscule point at the black hole's center.[7] This makes the center of a black hole both enormously massive and incredibly tiny, and hence it falls on both sides of the purported divide: we need to use general relativity because the large mass creates a substantial gravitational field, and we also need to use quantum mechanics because all the mass is squeezed to a tiny size. But in combination, the equations break down, so no one has been able to determine what happens right at the center of a black hole.

That's a good example, but if you're a real skeptic, you might still wonder whether this is something that should keep anyone up at night. Since we can't see inside a black hole unless we jump in, and, moreover, were we to jump in we wouldn't be able to report our observations back to the outside world, our incomplete understanding of the black hole's interior may not strike you as particularly worrisome. For physicists, though, the existence of a realm in which the known laws of physics